Links in Linux – Hard Link & Soft Link

Complete Notes with Inode Examples

1. Hard Link

A Hard Link is another name for the same file (points directly to the inode).
Create a Hard Link:

```
ln original.txt hardlink.txt
```

Example:

```
echo "Hello Linux" > original.txt
ln original.txt hardlink.txt
ls -li
```

Output:

```
12345 -rw-r--r-- 2 prashant users 12 Sep 22 20:10 hardlink.txt
12345 -rw-r--r-- 2 prashant users 12 Sep 22 20:10 original.txt
```

■ Both files share the same inode number and data.
Editing through hard link:

```
echo "New Line Added" >> hardlink.txt
cat original.txt
```

■ Changes are reflected in both because they share the same inode.

2. Soft Link (Symbolic Link)

A Soft Link is like a shortcut. It points to the path of the original file.
Create a Soft Link:

```
ln -s original.txt softlink.txt
```

Example:

```
echo "Linux World" > original.txt
ln -s original.txt softlink.txt
ls -l
```

Output:

```
lrwxrwxrwx 1 prashant users 12 Sep 22 20:15 softlink.txt -> original.txt
```

■ Symlink points to the path, not the data. If the original is deleted, the symlink breaks.

3. Difference Between Hard Link and Soft Link

Feature	Hard Link	Soft Link (Symlink)
Inode Number	Same as original	Different (points to path)
Works if file deleted?	Yes	No, becomes broken
Across filesystems	Not allowed	Allowed
Points to	Actual data (inode)	File path

4. Useful Commands

List inode numbers:

```
ls -li
```

Follow symlink to target:

```
readlink softlink.txt
```

Remove a link:

```
rm hardlink.txt
```

```
rm softlink.txt
```

Summary

- Use `ln file linkname` → Hard link
- Use `ln -s file linkname` → Soft (symbolic) link